How it works

This document describes the logic inside `RCM builder.html` so a reviewer can follow what the tool does without reading the code.

What the tool does

`RCM builder.html` is a single-page, browser-based builder for a draft Risk and Control Matrix (RCM). You paste or upload a process narrative (.docx, .xlsx, .txt or .csv), and it turns the narrative into a table of risks and controls, with built-in format checks. It was built and tested around SOX / internal-control process narratives (for example Fixed Assets), but it works on any narrative written as full sentences.

In short, the tool:

- reads the narrative and loads it into an editable text box;
- drafts a baseline RCM with one row for each control-like sentence, leaving anything the narrative does not state blank or flagged instead of guessing it;
- builds a structured AI prompt (role, output structure, seven rules) that you run in ChatGPT, Claude or Gemini;
- imports the AI's table or JSON reply, skips look-alike duplicates, fills blank cells and keeps the AI's FLAGS line;
- checks every row for form (Key / Non-Key value, one approved assertion, a non-generic test procedure, duplicates) and highlights gaps;
- lets you edit, add or delete rows, and exports the RCM to Excel, Word or CSV.

Everything runs in the browser, so the narrative and the RCM never leave your machine. The only step outside the tool is the one you do yourself: pasting the prompt into your own AI tool.

1. Pipeline

```
Narrative (paste / upload) -> Baseline RCM (optional, rule-based)
   -> Build AI prompt -> Run in your own AI -> Paste reply
   -> Import (de-duplicate, fill blanks, keep FLAGS) -> Validate
   -> Review / edit -> Export (Excel, Word, CSV)
```

2. Reading the narrative

- Accepts .docx, .xlsx, .txt, .csv. Anything else shows an "Unsupported file type" message (for older .doc or .xls files, save as .docx or .xlsx first, or paste the text).
- .txt and .csv files are read as plain text, exactly as they are. A CSV is not split into columns.
- .docx: the text of every paragraph in the document body becomes one line.
- .xlsx: every sheet is read; each non-empty row becomes one line, with its non-empty cells joined by - and a full stop added at the end.
- .docx and .xlsx files are unzipped inside the browser with built-in functions; no library is used.
- If no text can be read (for example a scan or an image), a message asks you to paste the text instead.
- The text lands in the Narrative box, where you can correct it before generating anything. The process name (optional, 60 characters) is used for the export file name and the Word title.

3. Baseline RCM (step 1)

The baseline is a rule-based first draft, with no AI. Whitespace is collapsed and the narrative is split into sentences after a full stop, question mark or exclamation mark. A sentence becomes a draft control when it:

- is at least 15 characters long;
- contains a control word (see below);
- is not identical to a control already in the table.

Control words (matched at the start of a word, so *approves* and *approved* both count): approv, authoris / authoriz, review, reconcil, verif, check, match, validat, sign off / signed off, inspect, compar, confirm, monitor, segregat, block, limit, dual, escalat.

Column	How the baseline fills it
Sr. No.	Numbered 1, 2, 3 ... in table order. Renumbered automatically after every change.
Process Step Ref.	"Step N", where N is the position of the sentence in the narrative (a sentence count, not the narrative's own numbering).
Risk	Left blank. The narrative rarely states the risk.
Control Description	The sentence itself.
Control Owner	The first job title found in the sentence from a fixed list (for example *finance controller*, *accounts payable team*, *procurement team*, *warehouse manager*, *HR manager*, *payroll team*, *CFO*, *manager*, *director*, *head*, *requester*, *approver*, *clerk*, *supervisor*). If none is found: "Owner not specified in narrative".
Frequency	The first match in this order: *daily*, *weekly*, *month-end / monthly* (Monthly), *quarterly*, *annual / yearly* (Annual), then *each*, *every*, *per*, *upon*, *whenever*, *before*, *prior to* (Per transaction). If none: "Not specified".
Key / Non-Key, Test Procedure, Assertion	Left blank. These need judgement, so they are completed by the AI step or by the reviewer.

A message reports how many controls were drafted. If none are found, the tool asks for full sentences, one step per sentence, using words such as approves, reviews or reconciles.

4. AI prompt (step 2)

Build AI prompt creates the prompt in the browser and shows it in an editable box, with the narrative inserted between ### markers. The tool never sends it anywhere; you copy it (Copy prompt button) into ChatGPT, Claude or Gemini.

Prompt section	What it says
ROLE	Act as an RCM generator: an internal-audit specialist assistant that converts a process narrative into an RCM suitable for a SOX / internal-control review.
OUTPUT STRUCTURE	Nine columns in a fixed order: Sr. No., Process Step Ref., Risk, Control Description, Control Owner, Frequency, Key / Non-Key, Test Procedure, Assertion. One row for each control. A FLAGS line at the beginning summarising gaps. A line at the end with the total number of controls, split Key / Non-Key.
RULES	Seven rules (table below).

Prompt section	What it says
NARRATIVE	The narrative text.

#	Rule in the prompt
1	Use only the narrative. Do not invent risks, controls, owners, thresholds or frequencies that are not stated or directly implied.
2	If the owner is not named, write "Owner not specified in narrative" and add the row to FLAGS.
3	If the frequency is not stated and cannot be reasonably inferred (annual, monthly, per transaction), write "Not specified" and add it to FLAGS.
4	Classify each control as Key or Non-Key by judgement of how material the risk is if the control fails.
5	Do not merge two distinct controls into one row, and do not split one control into two rows.
6	Number rows sequentially, in the order the controls appear in the narrative.
7	If the narrative is silent on a control a reviewer would expect (for example segregation of duties), do not fabricate one. Note the gap in FLAGS.

The Assertion column allows one primary assertion per control, from eight approved values: Occurrence, Completeness, Accuracy, Cutoff, Existence, Valuation, Rights and Obligations, Presentation and Disclosure.

If you edit the prompt and then click Build AI prompt again, the tool asks you to confirm before it replaces your edits.

5. Importing the AI reply

Paste the AI's reply and click Import AI rows. Three layouts are accepted: a markdown (pipe) table, a tab-separated table, or a JSON array of objects.

- Code fences, bold markers (`**`) and `<br>` tags are stripped first.
- **JSON:** each key that matches one of the nine columns is mapped to it. Any other key becomes a new extra column.
- **Table:** if a header row is found (every cell 40 characters or less, and at least four cells matching known column names), columns are read by name. Without one, the default column order is used. Unrecognised extra headers become new columns.
- A line starting with FLAGS is stored as the AI's flag text; following lines that are not table rows are added to it.
- A line starting with Total is read for its first number, kept as the AI's stated control count.
- Rows with neither a risk nor a control are skipped.
- New rows are appended at the end of the table, after any existing rows.
- Flags from each import are kept as "AI reply 1: ...", "AI reply 2: ...". The same flag text is never added twice.
- If the AI's stated total differs from the number of rows in the table, the validation line says so.
- A message reports new rows, new columns, rows recognised as existing, blank cells filled and empty rows skipped.

6. Duplicate matching

Before an imported row is added, it is compared with the rows already in the table, so that importing after the baseline (or importing the same reply twice) does not double up the controls.

- **Similarity** is the share of words two texts have in common (shared words divided by all distinct words), after dropping common words (the, of, and, to ...) and trimming endings such as -ation, -ing, -ed, -es, -ion, -al, -s from longer words.

Situation	Rule for "looks like an existing row"
Both rows have a control description	Control text at least 60% similar, or at least 50% similar with risk text also at least 50% similar.
Either control is blank or a placeholder (blank, "not stated", "not specified", "owner not specified", "no control identified")	Risk text at least 60% similar.

- A matched row is **not added**. Instead, any blank or placeholder cell in the existing row (every column except Sr. No. and Control Description) is filled from the incoming row. Real values already in the table are never overwritten.
- The closest match wins, and each existing row can be matched only once in a single import.
- Tick "Add every row, even if it looks like an existing one" to switch matching off and import every row as new.

7. Validation

Every row is checked each time the table changes. The checks look at form only.

Column	Check	Message shown
Risk	Not blank	Risk is blank
Control Description	Not blank. Not a duplicate of another row (same text after ignoring case, punctuation and spacing; "no control identified" is ignored).	Control is blank; Duplicate of row N
Control Owner	Not blank	Owner is blank (use "Owner not specified in narrative" if not stated)
Frequency	Not blank	Frequency is blank (use "Not specified" if not stated)
Key / Non-Key	Exactly Key or Non-Key (Non Key and NonKey are also accepted)	Must be exactly Key or Non-Key
Assertion	Exactly one of the eight approved assertions. "Cut-off" and "&" are read as Cutoff and "and".	Assertion is blank; More than one assertion given (pick one primary); Not one of the eight approved assertions
Test Procedure	Not blank, at least 20 characters, and not just a bare "review", "check", "test", "verify" or "inspect"	Test procedure is blank; Test procedure is too generic (say what is tested and on what sample)

- **Red cell:** the cell fails a check. Hover over it to see the reason.
- **Orange cell:** Risk, Control, Owner or Frequency is blank or holds a placeholder such as "Not specified". It is a gap the narrative did not state. It passes the format check but is highlighted so it is not missed. Red takes priority over orange.
- **Summary line:** "Validation: X of Y rows pass. Z need review." Up to 40 issues are listed beneath it as "Row N, Column: message", followed by a count of any more.
- **Count line:** "N controls (K Key, M Non-Key)". A value containing "non" counts as Non-Key, any other value containing "key" counts as Key, and a blank counts as neither.
- Validation cannot judge whether a risk is well worded, whether a control is truly Key, or whether the assertion is the right one. Those stay with the reviewer.

8. Editing the table

- Click any cell to edit it. The table is re-checked when the cell loses focus.
- Add row adds an empty row. The × button deletes one row. Clear all rows empties the table and the stored FLAGS (after a confirmation).
- Clear inputs (step 1) and Clear prompt and reply (step 2) empty those boxes only. Clear everything (steps 1 to 3) empties all three steps after a confirmation.
- Extra columns created by an AI reply appear in the table and are included in every export.

9. Export

Format	Content
Excel (.xlsx)	Sheet "RCM": a header row (white bold text on navy, frozen) and one row per control, with wrapped text and thin borders. Sheet "Review notes": the FLAGS text, total controls with the Key / Non-Key split, the validation result, and the line "Draft for professional review."
Word (.docx)	A4 landscape. Title "Risk and Control Matrix - <process> (draft)", the FLAGS paragraph, the table (navy header row repeated on each page, rows kept whole across pages), total controls, the validation result and the draft note.
CSV (.csv)	UTF-8 with a byte-order mark so Excel opens it correctly. Header row, and every value in quotes.

- The file name is the process name with symbols replaced by underscores, plus `_RCM` (default `Process_RCM`). With no rows, the tool shows "Nothing to export yet."
- Exports contain the table as it stands, including blank or flagged cells. The red and orange highlighting is not carried into the files.
- **Formula protection:** in the CSV, any value starting with `=`, `+`, `-`, `@`, a tab or a carriage return gets an apostrophe in front. In the Excel file every value is written as text, and in the Word file as plain text, so neither can run as a formula. Control characters are removed from both.

10. Privacy and dependencies

- The tool makes no network requests and loads no libraries. It is plain HTML, CSS and JavaScript in one file, and works offline.

- Nothing is stored. The tool starts blank on every load, and removes any rows an older version left in browser storage.
- Excel and Word exports are written, and .docx and .xlsx narratives are read, directly as zipped XML, with no library.
- The AI step is the one place data can leave your machine, and only because you paste the prompt into a third-party tool yourself. Use an AI tool approved for the data, and remove client identifiers from the narrative first.

Browser feature	Used for
File reader, Blob download	Loading the narrative file and downloading the exports
DecompressionStream, DOMParser	Unzipping and reading .docx and .xlsx narratives
Clipboard	Copy prompt button (falls back to the older copy command)
prefers-color-scheme	Dark mode: colours adapt to the operating system setting

11. Known limitations

- **The baseline is a rough scaffold on long narratives.** It is keyword-based. It can miss controls that use other wording, treat a heading as a control, or split one control across two rows. Step numbers count sentences, not the narrative's own numbering. Use the AI step for full coverage.
- **Sentence splitting** happens after every full stop, question mark or exclamation mark followed by a space, so an abbreviation such as "e.g." or "No." can cut a sentence early.
- **The owner detector uses a fixed list of job titles.** It can capture part of a title ("Head" from "Plant Head"), pick up a wrong word, or show an owner as unspecified when the title is not on the list ("Stores team").
- **Rows imported from an AI reply are appended at the end,** so row order can differ from the narrative. Process Step Ref. keeps the link.
- **Validation checks form only.**
- **The Excel export holds up to 26 columns** (A to Z): the nine standard columns plus up to 17 extra ones.
- The output is a draft for professional review, not an audit conclusion.

12. Manual test cases

Tests 1 to 7 use the files in `demo/fixed-assets-quick-example`.

#	Test	Expected
1	Upload Fixed_Assets_Narrative.docx	Narrative box fills; message "Loaded Fixed_Assets_Narrative.docx (1086 characters)"
2	Click Generate baseline RCM	8 controls drafted; Risk, Key / Non-Key, Test Procedure and Assertion blank; validation "0 of 8 rows pass"
3	Click Build AI prompt	Prompt in step 2 matches Fixed_Assets_AI_Prompt.txt

#	Test	Expected
4	Paste Fixed_Assets_Sample_AI_Reply.md and click Import AI rows	1 new row; 8 rows recognised as existing and 33 blank cells filled; FLAGS shown below the table; "9 controls (7 Key, 2 Non-Key)"; validation "9 of 9 rows pass"
5	Click Import AI rows again with the same reply	0 new rows; 9 rows recognised as existing; table unchanged
6	Tick "Add every row" and import again	9 more rows added (18 controls)
7	Change the owner of row 7 from "Head" to "Plant Head"	Cell updates; the row still passes validation
8	Import a JSON reply with a Comment key and the assertion "Accuracy & Cut-off"	1 new row and 1 new column (Comment); assertion cell red with "More than one assertion given"
9	Export CSV with a control that starts with =	The value starts with an apostrophe in the file
10	Upload a .pdf file	"Unsupported file type" message
11	Click Generate baseline with an empty box, then with a sentence that has no control word	"Paste or upload a narrative first", then "No controls found" message
12	Click Export Excel with no rows; then with rows and the process name "Fixed Assets"	"Nothing to export yet"; then Fixed_Assets_RCM.xlsx, .docx or .csv downloads
13	Click Clear everything and confirm	Narrative, prompt, reply and table are all empty
14	Switch the OS to dark mode	Colours adapt